

Kylee McCarthy, MS

Director of Technology Operations

Saratoga Springs, UT (Remote) · kyleelinmccarthy@gmail.com · kyleemccarthy.com · linkedin.com/in/kylee-mccarthy-706499123

Summary

I taught myself to code by automating my own job, and now lead a 20-person technology department. I still write production code, so the standards and tooling I put in place fit how the work actually happens — delivery is five times what it was, with fewer people and the highest satisfaction the department has recorded. I'm looking to do this again somewhere technology is central to what the business does and the ambition matches what a team like this can build.

Professional Experience

National Benefit Services, LLC · Remote · 9 years in technology, 11 with the company

Director of Technology Operations · Nov 2022 – Present

- Grew delivery from about 40 backlog items a year to 200, with a smaller team each year. Enterprise projects went from 1 completed in 2023 to 14 today — 6 in production, 4 releasing, 4 in active development. I built 11 of the 14 myself.
- Built the engineering operating model the department runs on: a design system, a project template every codebase starts from, and a versioned standards plugin that enforces itself in CI. Documentation is part of the definition of done, so decisions outlast the people who made them.
- Brought the department onto AI-assisted development and built the review, testing and documentation standards that came with it — none of which existed before.
- Made the case to build rather than buy, then delivered it as the primary engineer. The project tracker, accessibility vendor and engagement tool we licensed are already retired and replaced; the recordkeeping platform is next.
- Help desk volume fell to its lowest recorded level and satisfaction rose to its highest, from business users and staff alike, while I moved newly promoted supervisors into running their own teams.

Information Systems Analyst Lead · Nov 2019 – Nov 2022

- Led a team of analysts across project pipelines and quality standards, and built the early documentation and governance frameworks the later department buildout was based on.

Information Systems Analyst · Jul 2017 – Nov 2019

- Earned a project lead role within a year, owning the full development lifecycle on C# MVC applications and replacing manual spreadsheet workflows with T-SQL stored procedures.

Assistant Account Manager · May 2016 – Jul 2017 | Benefits Specialist · Mar 2015 – May 2016

- Administered benefits for a portfolio of employer clients, then taught myself VB, JavaScript and REST APIs and automated the manual parts of the job without being asked to. The automation made the role redundant and leadership moved me into Information Systems.

Platform Portfolio

In production

- **403HQ — two client deployments**, ~10,000 employees each. Shipped the first in 6 weeks to keep a client that was leaving, then built a second version for another client with its own branding, plan rules and customizations.
- **AURA** — accessibility compliance as a single script tag. Replaced an \$800/year two-site vendor license; now on 15+ sites at no extra cost.
- **Beacon** — replaced Azure DevOps and pulled GitHub Enterprise and ServiceDesk Plus into one project model. Shapes work before it starts, reviews every pull request against our standards with AI, and feeds the results to performance scorecards. Also carries the technology asset inventory with dependency mapping and compliance scoring, generates SOC audit evidence packs per project, and scans every site for ADA compliance.
- **Ruckus** — ten browser-based multiplayer games with live polls and quizzes over SignalR, for a company that had gone remote-only. Built in weeks; replaced AhaSlides.

Releasing

- **NBS Website** — replaced a dated WordPress site with Next.js and .NET across 66 pages, rebuilt around participants, sponsors and advisors, with the CMS marketing had wanted for years.
- **Online Forms, Secure Upload, Onboarding** — multi-tenant services with headless packages, so the website and Paragon share one implementation instead of three.

Building

- **External Payroll Processing** — ingests employer payroll files with no per-employer setup, validates against plan rules, resolves exceptions and exports to recordkeepers.
- **Paragon** — replacing FIS Relius for 403(b), 457 and FICA Alternative administration across 616 endpoints and 194 entities.

AI in Production

- Six systems run AI under audit across three providers. Documents are read as native PDF and image blocks and returned through forced tool-use against a JSON schema, so results are structured rather than parsed out of prose — no separate OCR stack to maintain.
- Every extraction carries a confidence score, and configurable auto-approve and manual-review thresholds decide whether work posts, queues for a human reviewer, or is rejected. Admins toggle extraction and background auto-processing at global, employer and plan scope, so AI can be turned off for one client without a deploy.
- Beacon reviews every pull request against those same standards automatically, so a model writing the first draft never means less scrutiny than a person writing it.

Personal Projects

- **Doing The Thing** — chore app I built for my own kids so they own their list instead of being nagged about it. Runs locally for my family.
- **Ember Tattoo & Piercing** ember-tattoo-web.vercel.app — studio site with booking, artists, gallery and aftercare, all editable through a CMS.
- **Family Budgeting Web App** — budgeting by pay period rather than calendar month, covering accounts, expenses, debts and school funds.
- **Kingdoms & Crowns** kingdomsandcrowns.com — homeschool hub that tracks a full week per child. Parents set each student's subjects, link their curriculum sites and build custom assignments; kids work through them on a timer that persists across tabs and devices, and earn rewards for staying on task. In beta with a small group of families.
- **Personal Portfolio Site** kyleemccarthy.com — statically rendered, with accessibility checked in CI by Playwright and axe.
- **ChemTree Games** chemtreegames.com — studio site for the games below.
- **ChemTreeHQ** — shared workspace for remote projects. Documents edit live through Yjs CRDTs, so simultaneous edits merge instead of overwriting.
- **Eliminated** — my first solo game project: a 3D multiplayer party game in Unity and C#, where a lobby is whittled down through a gauntlet of playground minigames until one player is left.
- **Eliminated (web)** eliminatedgame.com — the browser prototype that proved the concept before the 3D build. Twelve minigames with bot AI on an authoritative WebSocket server, so the server owns state and clients can't cheat.
- **The Wretched Few** — multiplayer roguelite in Unity, built with one friend and headed for Steam. You play the misunderstood monsters, hunted by humans who have been told they are evil. I own the story, the HUD and player UI/UX, and the art.

Skills

Leadership: Agile, Shape Up; department operations, SDLC governance, engineering standards and definition of done, technical documentation practice, roadmapping, hiring and supervisor development, vendor management, stakeholder communication, SOC audit readiness

Engineering: C#, .NET (every version since 2.0), TypeScript, Next.js, React, ASP.NET Core, MVC, EF Core; Clean Architecture, multi-tenancy, OAuth 2.0, OpenID Connect, step-up MFA

Databases: SQL Server, Azure SQL, PostgreSQL, MySQL, Oracle, Redis; schema design, stored procedures, migrations, tuning, backup and recovery. No separate DBA — database administration is mine.

Azure and infrastructure: App Service, Key Vault, Entra ID, Entra External ID, Managed Identity, Blob Storage, Application Insights, Container Apps, Bicep, Terraform, GitHub Actions, Docker, Railway, Vercel; CI/CD pipelines with contract and documentation gates, asset inventory and dependency tracking. I own our Azure environment end to end.

AI: Anthropic Claude, OpenAI, Azure OpenAI, Semantic Kernel; structured tool-use output, document and vision extraction, confidence thresholds, per-tenant governance, caching and cost control

Design: Figma, Adobe Creative Suite; design systems, component libraries, light and dark theming, WCAG 2.1 AA

Education

M.S. Technology Management, Western Governors University, 2022

B.S. Marketing Management, Western Governors University, 2020